

1. Determine which amounts of postage can be formed using three-cent and five-cent stamps. (For example, eight cents can be formed; two cents cannot be.) Prove this using strong induction.

Solution: We can form 3, 5, 6, and any number greater than or equal to 8. Proof: we can form 8, 9, and 10, and, for any number k , we can form it if we can form $k - 3$.

2. Let $\phi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Let f_n be the n -th Fibonacci number, starting $f_1 = 1, f_2 = 1, f_3 = 2$. Prove that $f_n = \frac{1}{\sqrt{5}}(\phi^n - \psi^n)$.

Solution: The base case ($n = 1$ and $n = 2$) we can verify directly. For the inductive step, we write

$$\begin{aligned} f_{n+1} &= f_{n-1} + f_n \\ &= \frac{1}{\sqrt{5}}(\phi^{n-1}(1 + \phi) + \psi^{n-1}(1 + \psi)). \end{aligned}$$

We now just check that $1 + \phi = \phi^2$ and similarly for ψ .

3. Once again, let f_n be the n -th Fibonacci number. Prove that $f_{n+1}f_{n-1} - f_n^2 = (-1)^n$ for all $n \geq 1$. (We say that $f_0 = 0$.)

Solution: We verify this in the case $n = 1$ and $n = 2$. For induction,

$$\begin{aligned} f_{n+2}f_n - f_{n+1}^2 &= (f_{n+1} + f_n)(f_n) - f_{n+1}(f_n + f_{n-1}) \\ &= f_{n+1}f_n + f_n^2 - f_{n+1}f_n - f_{n+1}f_{n-1} \\ &= f_n^2 - f_{n+1}f_{n-1} \\ &= -(-1)^n. \end{aligned}$$

4. Let g be a recursive function defined by $g(k) = 1 - g(k - 1)$. Prove that, for any value of $g(0)$, there exists some M such that $|g(k)| < M$ for all k . That is to say, no matter where you start, g does not grow infinitely.

Solution: We note that $g(k) = 1 - g(k - 1) = 1 - (1 - g(k - 2)) = g(k - 2)$. Therefore, we just have to take $M > \max(|g(0)|, |g(1)|)$.